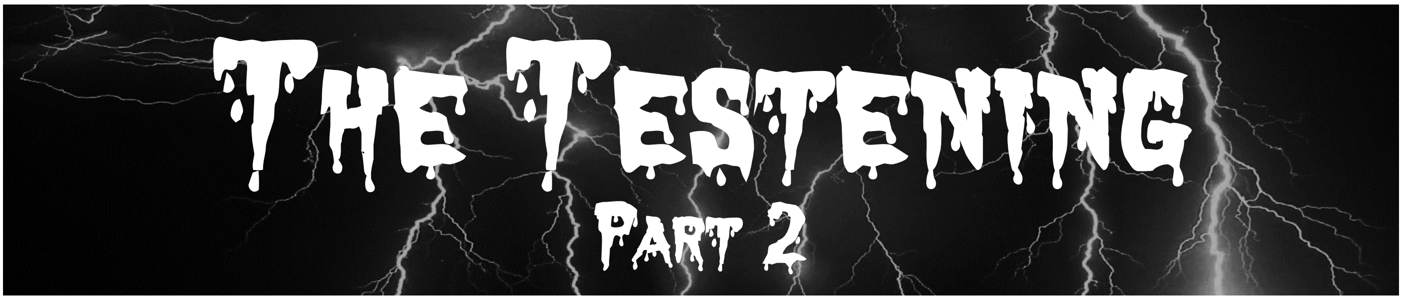

EEE3094S 2020

Information

- The test is open book.
- The test is out of 50, but there are 55 marks available in the paper. There are five questions.
- Submit your answers in pdf format. They may be typed or handwritten. Sketches may be hand-drawn and scanned, or constructed in a program such as PowerPoint.
- You must explain your reasoning and show your calculations for all questions.
- "Your mom" jokes date back at least 3500 years, with the oldest example found on a Babylonian tablet.
- The world's oldest ham is 119 years old. It is, allegedly, still edible.

Question 1

The frequency responses of three second-order systems are shown on the Nichols chart in Figure 1.A. They are all stable in open loop.

A) [3 marks] Are all of the systems stable in closed loop? Explain.

B) Rank the systems according to how they perform in feedback in each of the following criteria:
   i) [3 marks] Accurate position tracking (most accurate to least accurate)
   ii) [3 marks] Damping (most damping to least damping)
   iii) [3 marks] Speed (fastest to slowest)

C) [3 marks] All three of these systems meet the design requirements for an application, and you must select one to implement as a digital controller. The sampling rate has not been specified. Which one would you recommend? Why?

Total: 15 marks

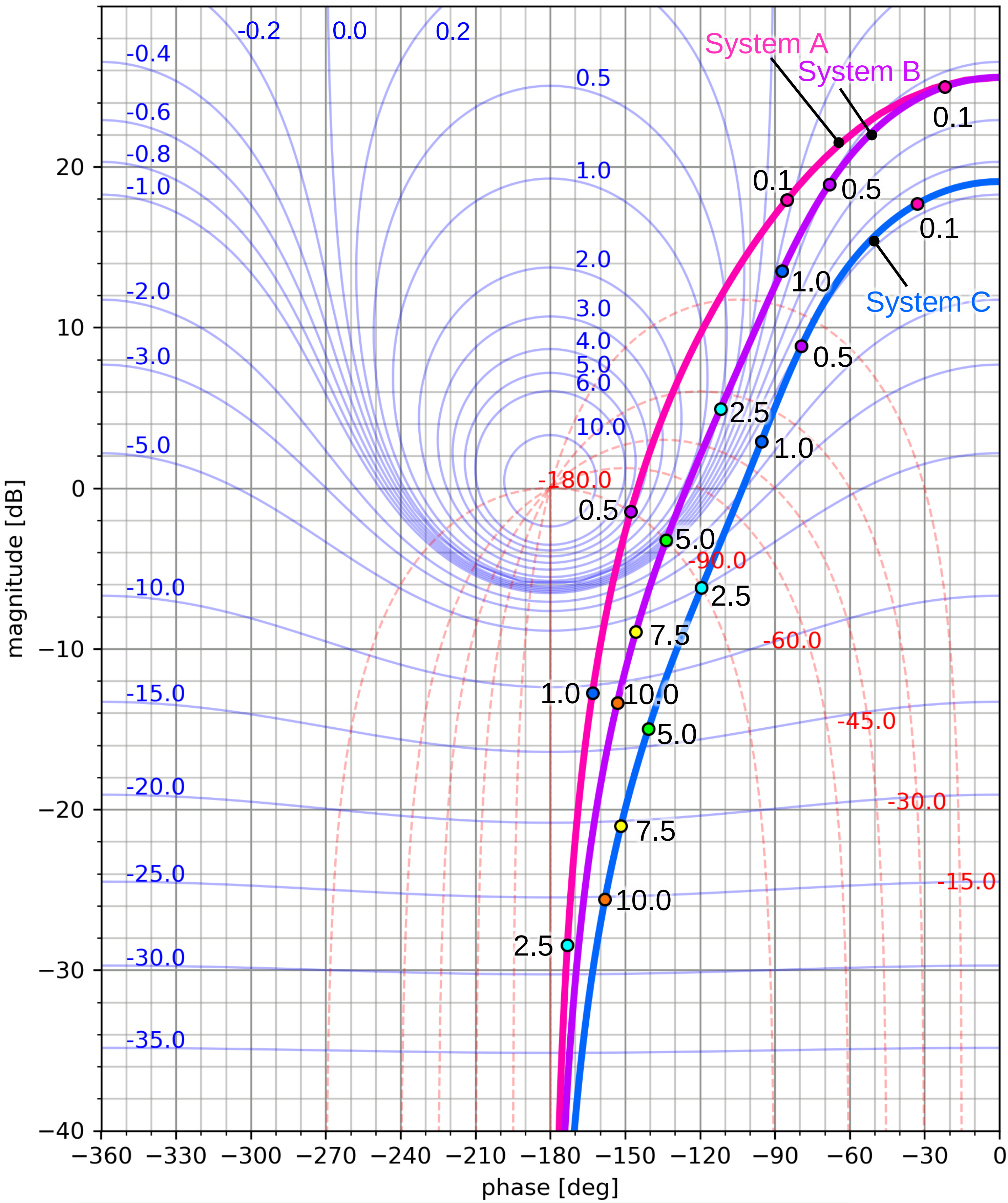

Figure 1.A: Nichols chart comparing systems A, B and C. Labelled frequencies are in radians per second.

Question 2

A) [6 marks] A first-order system produces the output shown in figure 2.A in response to a unit step. Identify a transfer function for this system. You may model the dead time using a first-order Pade approximation.

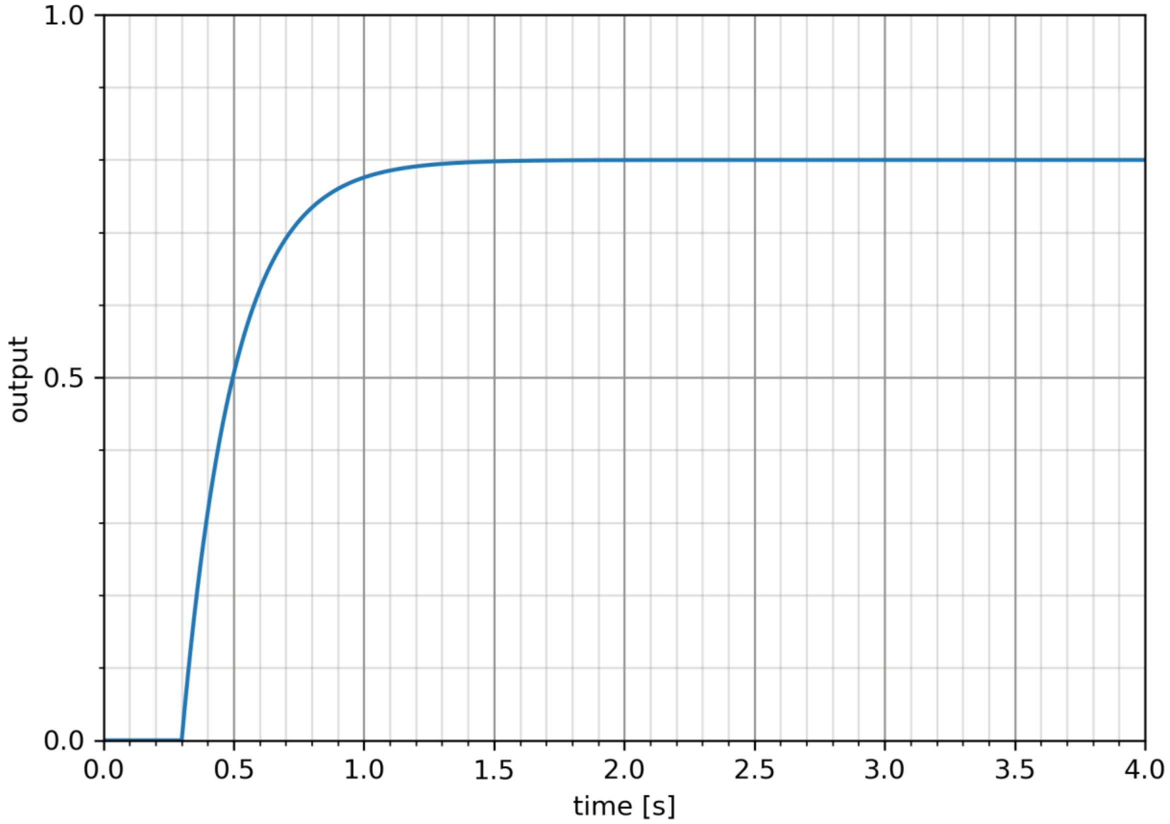

Figure 2.A: unit step response

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B) [4 marks] The Nyquist plot shown in figure 2.B shows the frequency response of a stable system. Will the system remain stable if it experiences a delay of 0.2 seconds?

Total: 10 marks

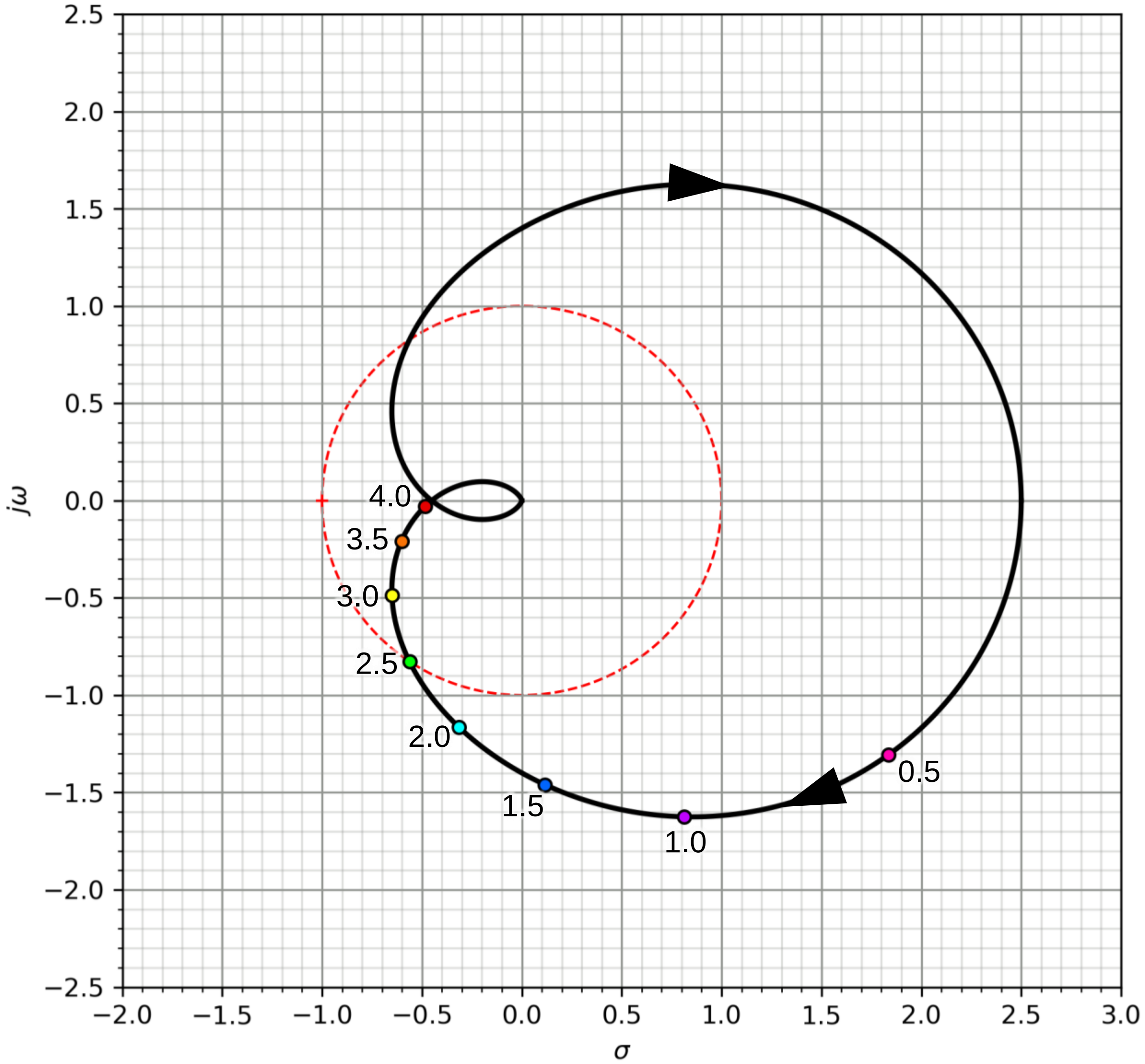

Figure 2.B: Nyquist plot. The labelled frequencies are in radians per second.

Question 3

A system has poor steady-state tracking accuracy, and it is determined that it can be sufficiently improved by increasing its gain by a factor of 10. Figure 3.A shows the system's frequency response on an inverse Nichols chart.

A) A controller with the frequency response shown in figure 3.B is suggested for the job.
   i) [1 mark] Is this a lead compensator or a lag compensator?
   ii) [4 marks] Would you recommend using this design? If not, suggest how it could be modified to make it suitable.

B) [4 marks] Would you suggest a proportional controller as an alternative? Explain.

Total: 9 marks

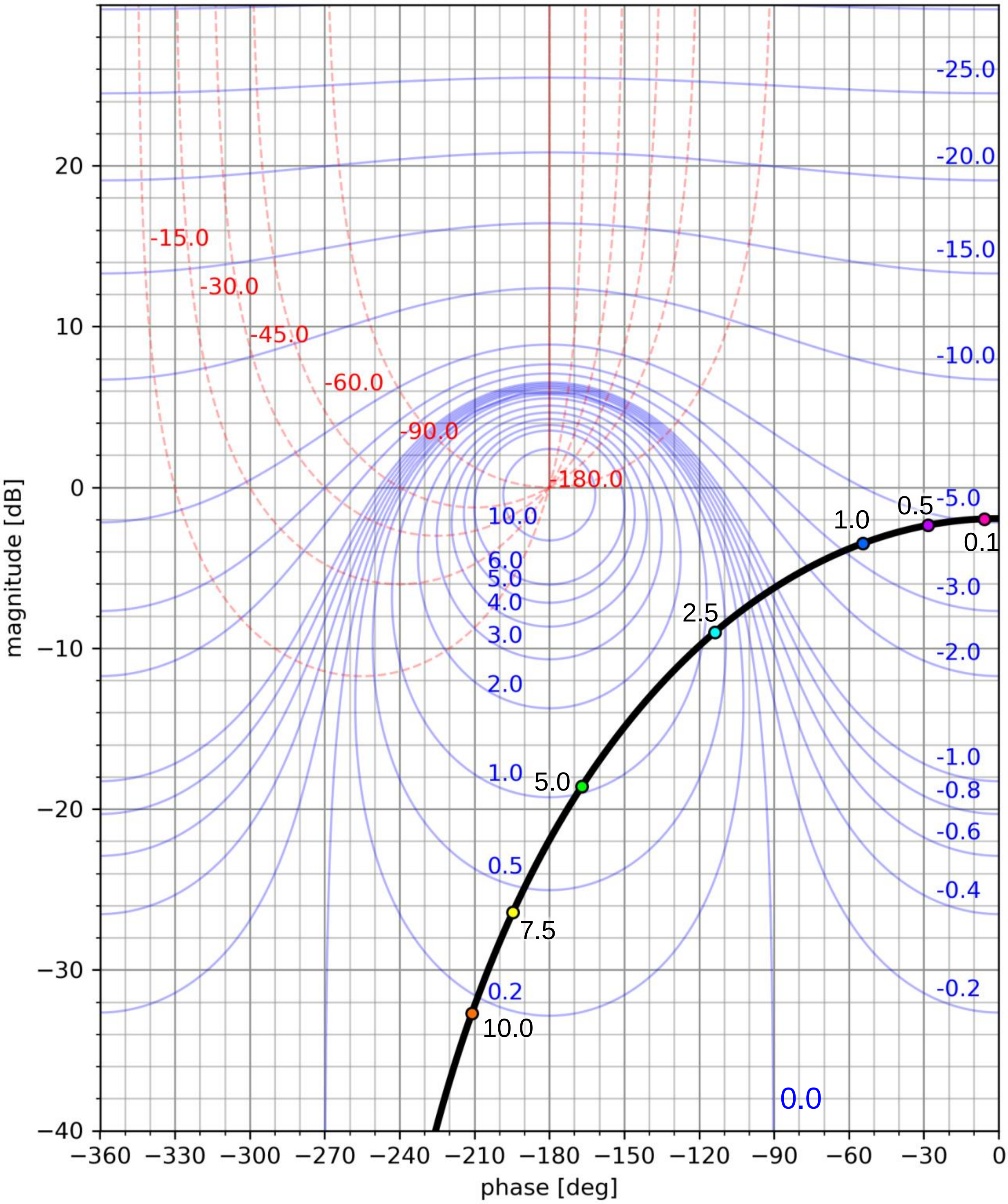

Figure 3.A: Inverse Nichols chart of uncompensated plant

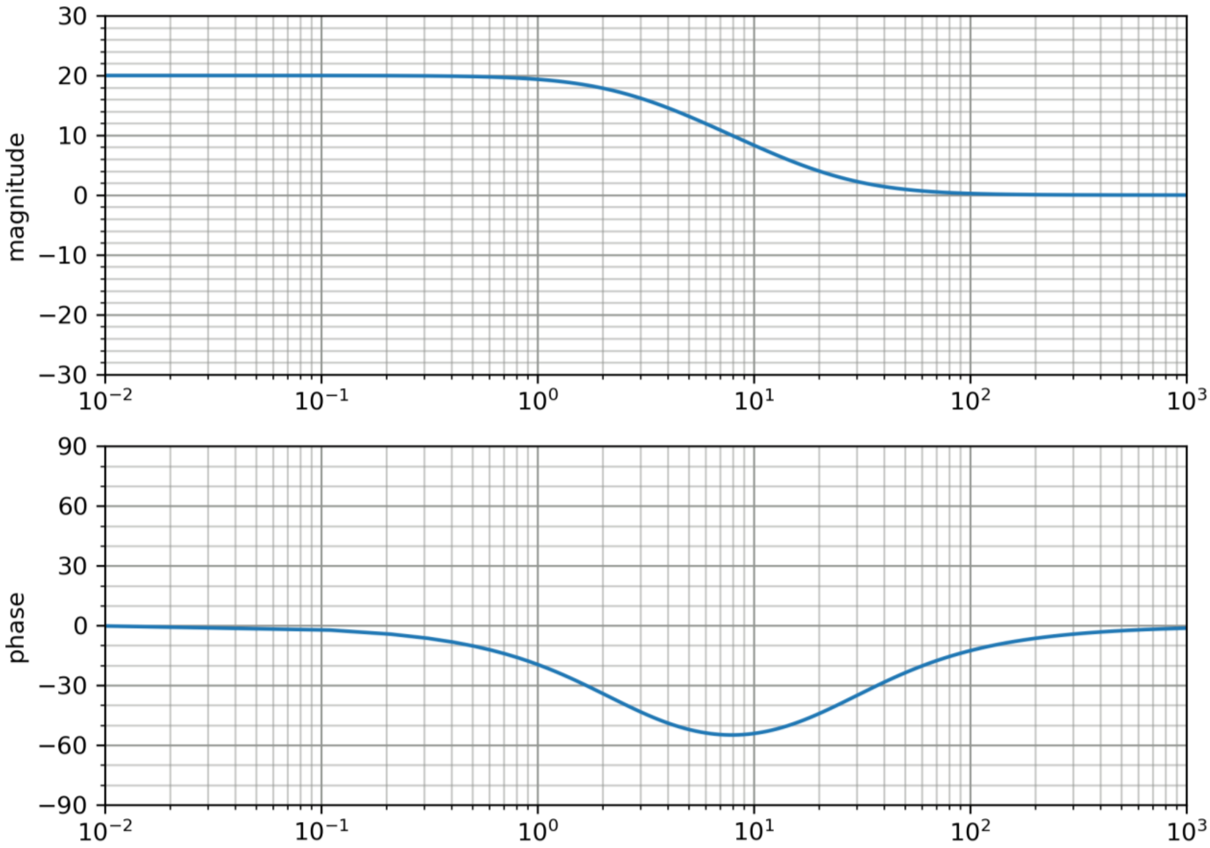

Figure 3.B: Bode plot of suggested controller

Question 4

Consider the Nichols chart shown in figure 4.A and answer the following TRUE or FALSE questions. Provide a short explanation for each answer. The system in the plot is stable in open and closed loop.

A) [2 marks] The system will not be destabilized by a delay.
B) [2 marks] The system will not be destabilized by increasing the gain.
C) [2 marks] The system can track a step setpoint with ≥90% accuracy.

Total: 6 marks

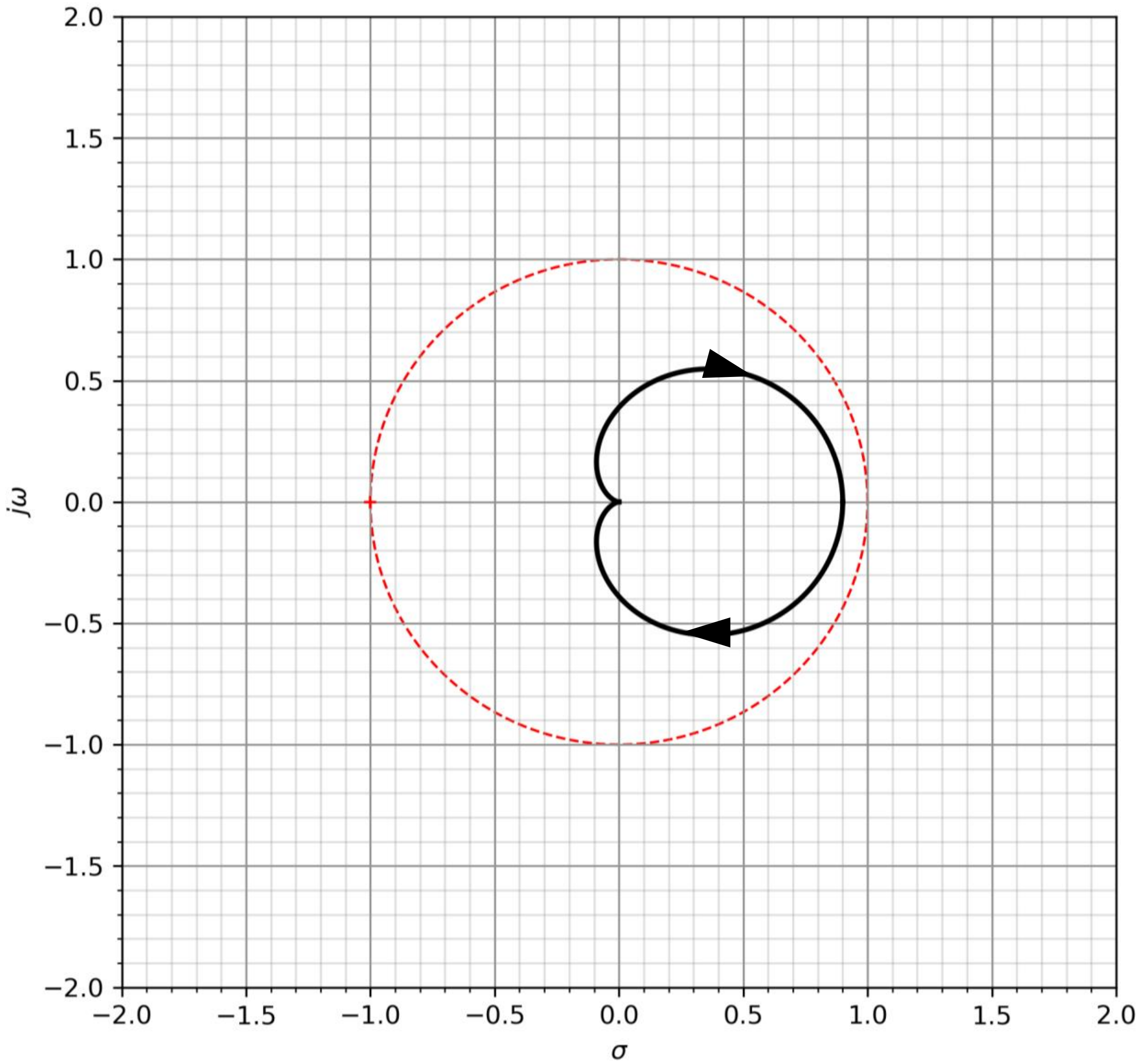

Figure 4.A: Nyquist plot

Question 5

A system has open-loop poles at -0.5 and -1. Its frequency response is shown in figure 5.A.

A) Decide whether the system satisfies the following design specifications in closed loop. Justify your answers.
   i) [2 marks] Phase margin of >60 degrees.
   ii) [2 marks] Percentage overshoot of less than 20%
   iii) [2 marks] Settling time to 2% of its final value in < 2 seconds.

Percentage overshoot: $100e^{-\frac{\zeta\pi}{\sqrt{1-\zeta^2}}}$

B) [9 marks] Design a lead compensator so the closed-loop system will satisfy the specifications in (A). There is no accuracy specification, but you should calculate the position error constant of your design.

Total: 15 marks

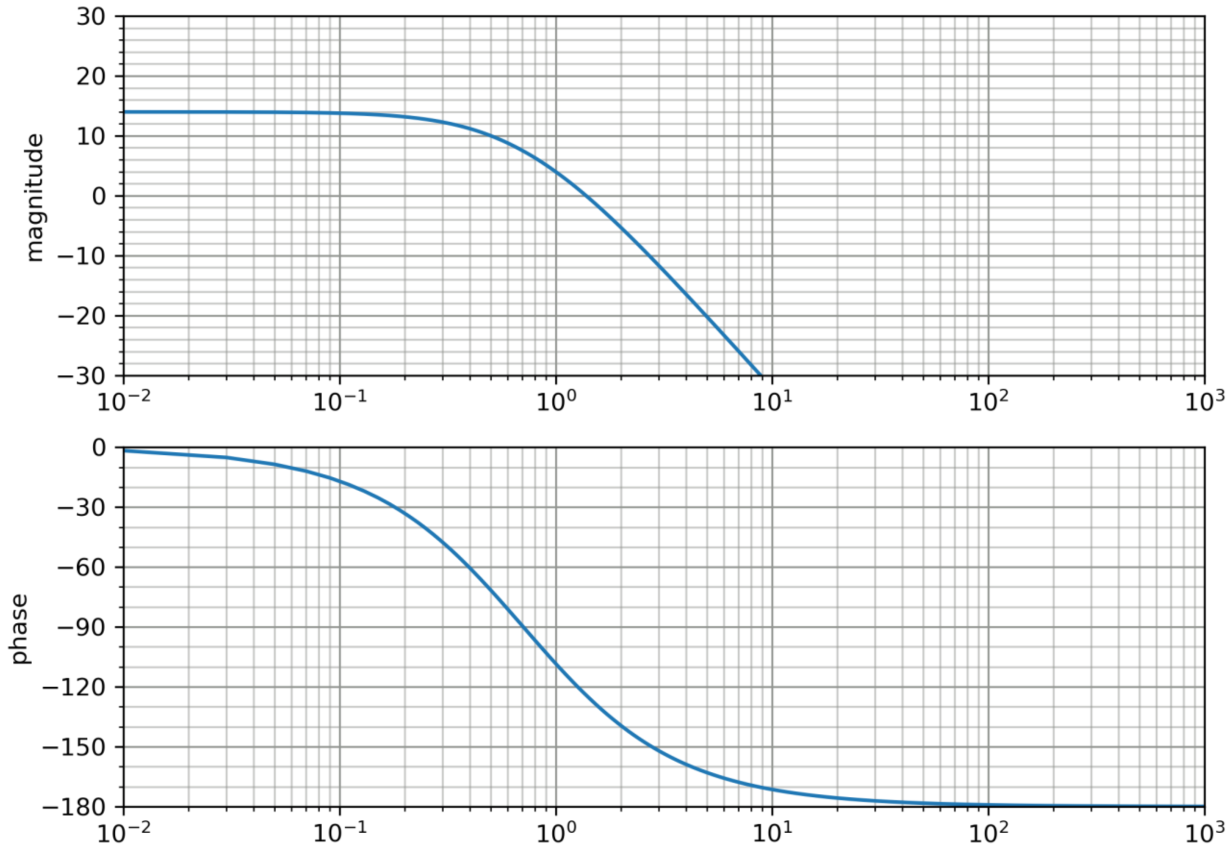

Figure 5.A: Bode plot